Convergence Tests

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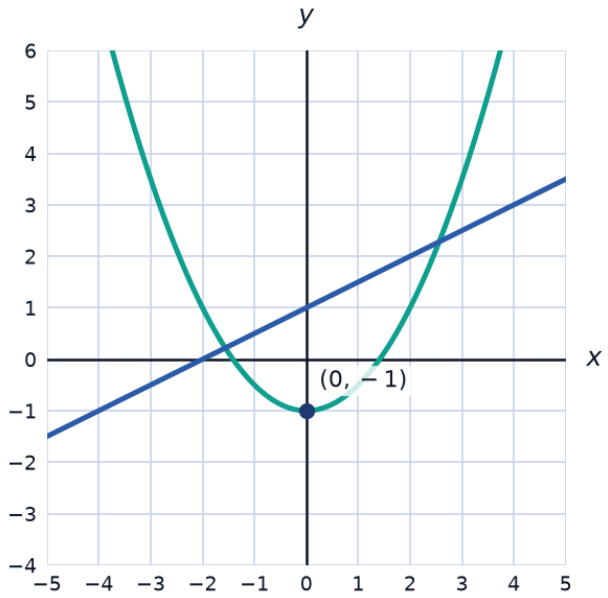
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: The ratio and comparison tests

Use the ratio test to determine whether $\sum_{n=1}^{\infty} \frac{n}{2^n}$ converges.

2 [Foundational] • Target: The ratio and comparison tests

Use the ratio test on $\sum_{n=1}^{\infty} \frac{n!}{n^n}$.



3 [Foundational] • Target: The ratio and comparison tests

Use the comparison test to determine whether $\sum_{n=1}^{\infty} \frac{1}{n^2+3}$ converges, by comparing to $\sum \frac{1}{n^2}$.

4 [Foundational] • Target: Alternating series and the p-series test

Determine whether each p-series converges or diverges:

a $\sum_{n=1}^{\infty} \frac{1}{n^3}$

b $\sum_{n=1}^{\infty} \frac{1}{\sqrt{n}}$

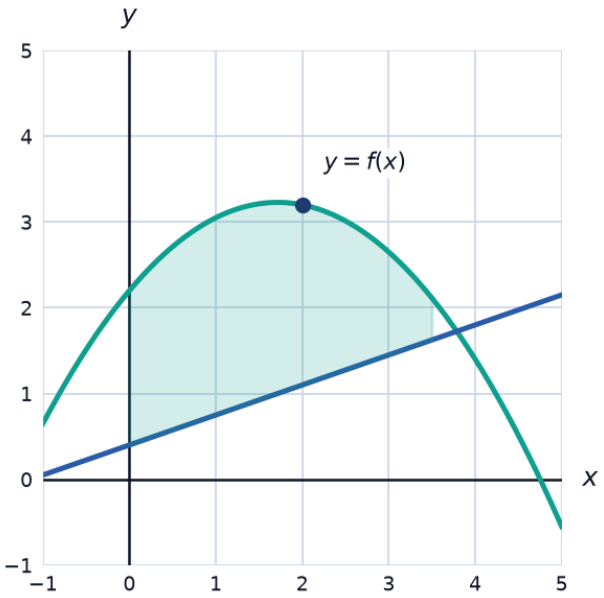
5 [Foundational] • Target: Alternating series and the p-series test

Determine whether $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n}$ converges, and classify it as absolutely convergent, conditionally convergent, or divergent.

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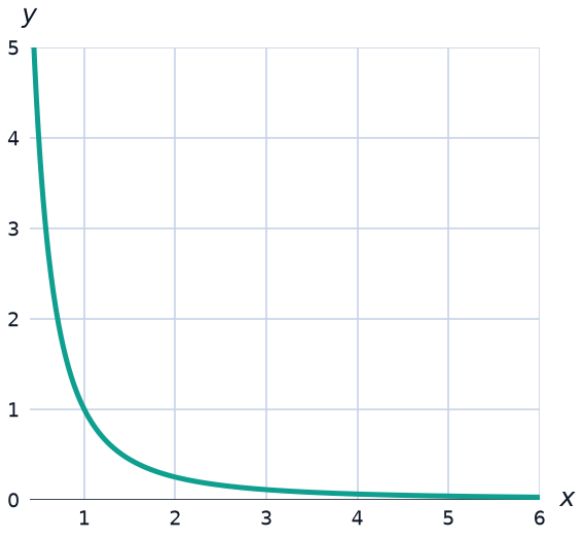
6 [Foundational] • Target: The integral test

Use the integral test to determine whether $\sum_{n=1}^{\infty} \frac{1}{n \ln n}$ (starting at $n = 2$) converges or diverges, using $\int_{2}^{\infty} \frac{1}{x \ln x}\,dx$.



7 [Foundational] • Target: The integral test

Use the integral test to confirm that $\sum_{n=1}^{\infty} \frac{1}{n^2}$ converges, using $\int_{1}^{\infty} \frac{1}{x^2}\,dx$.



Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

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8 [Intermediate] • Target: Choosing the right test

For each series, name the most efficient test to apply (ratio, comparison, p-series, or alternating series), without fully evaluating:

a $\sum \frac{2^n}{n!}$

b $\sum \frac{(-1)^n}{\sqrt{n}}$

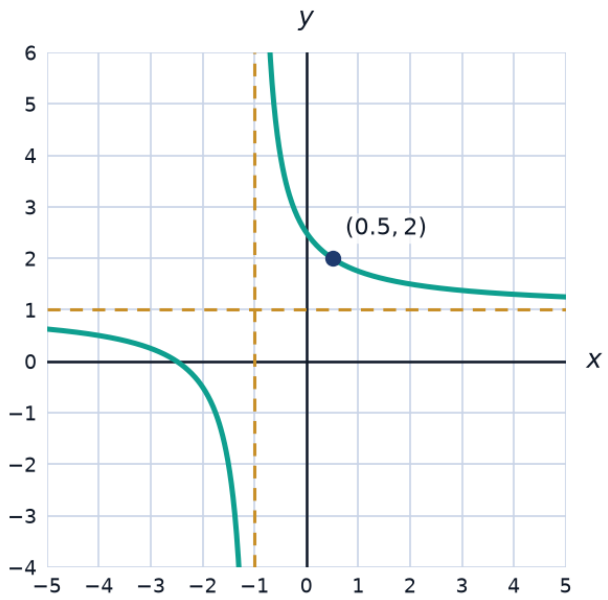
c $\sum \frac{1}{n^4}$

9 [Intermediate] • Target: The limit comparison test

Use the limit comparison test to determine whether $\sum_{n=1}^{\infty} \frac{n+1}{n^3+2}$ converges, comparing to $\sum \frac{1}{n^2}$.

10 [Intermediate] • Target: The limit comparison test

Use the limit comparison test on $\sum_{n=1}^{\infty} \frac{3n^2-1}{n^4+2n}$, comparing to $\sum \frac{1}{n^2}$.



11 [Intermediate] • Target: The root test

Use the root test to determine whether $\sum_{n=1}^{\infty} \left(\frac{n}{2n+1}\right)^n$ converges.

12 [Intermediate] • Target: The root test

Use the root test on $\sum_{n=1}^{\infty} \frac{3^n}{n^n}$.

13 [Intermediate] • Target: Absolute versus conditional convergence

Determine whether $\sum_{n=1}^{\infty} \frac{(-1)^n}{n^2}$ is absolutely convergent, conditionally convergent, or divergent.

14 [Intermediate] • Target: Absolute versus conditional convergence

Determine whether $\sum_{n=1}^{\infty} \frac{(-1)^n}{\sqrt{n}}$ is absolutely convergent, conditionally convergent, or divergent.

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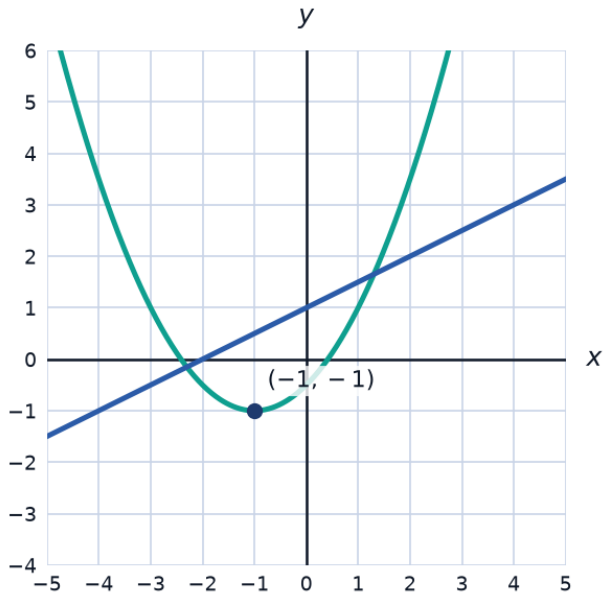
Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: Estimating the sum of an alternating series** • **Cross-Lesson: 7.01 Integration by Substitution**

The alternating series $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^2}$ converges by the alternating series test. Use the fact that the error after n terms is bounded by the first omitted term to estimate how many terms are needed for the partial sum to be within 0.001 of the true sum.

16 **[Challenging]** • **Target: A combined convergence-test challenge** • **Cross-Lesson: 7.02 Integration by Parts**

Determine whether $\sum_{n=1}^{\infty} \frac{n^2}{3^n}$ converges, using the ratio test.



17 **[Challenging]** • **Target: A combined convergence-test challenge** • **Cross-Lesson: 8.01 Area Between Curves**

Determine whether $\sum_{n=2}^{\infty} \frac{1}{n(\ln n)^2}$ converges, using the integral test with $u = \ln x$.

18 **[Challenging]** • **Target: A combined convergence-test challenge** • **Cross-Lesson: 8.02 Volumes of Revolution**

A student claims that since $\lim_{n\to\infty} \frac{1}{n} = 0$, the series $\sum_{n=1}^{\infty} \frac{1}{n}$ must converge. Explain what is wrong with this reasoning, and state the correct necessary (but not sufficient) condition that the nth-term test actually provides, using the harmonic series as a counterexample.

19 **[Challenging]** • **Target: A multi-step convergence investigation** • **Cross-Lesson: 8.03 Arc Length and Improper Integrals**

For the series $\sum_{n=1}^{\infty} \frac{n^3 + 2n}{2n^5 - n + 1}$, first find the nth-term limit to check the necessary condition, then use the limit comparison test with $\sum \frac{1}{n^2}$ to determine whether the series converges.

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20 [Challenging] • Target: A multi-step convergence investigation • Cross-Lesson: 9.01 Separable Differential Equations

A quality-control engineer models cumulative defect probability using $\sum_{n=1}^{\infty} \frac{1}{n \cdot 2^n}$. Use the ratio test to confirm this series converges, and briefly explain (in one sentence) why a convergent bound like this is useful for guaranteeing that a manufacturing process's total defect risk stays finite over infinitely many production runs.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Integral Test and p-Series

For what values of p does the p-series $\sum_{n=1}^{\infty} \frac{1}{n^p}$ converge?

a. $p > 1$

b. $p \geq 1$

c. $p < 1$

d. $p > 0$

22 [Foundational] • Target: Integral Test and p-Series

Use the Integral Test to determine the convergence of $\sum_{n=2}^{\infty} \frac{1}{n \ln n}$.

a. Diverges

b. Converges to 1

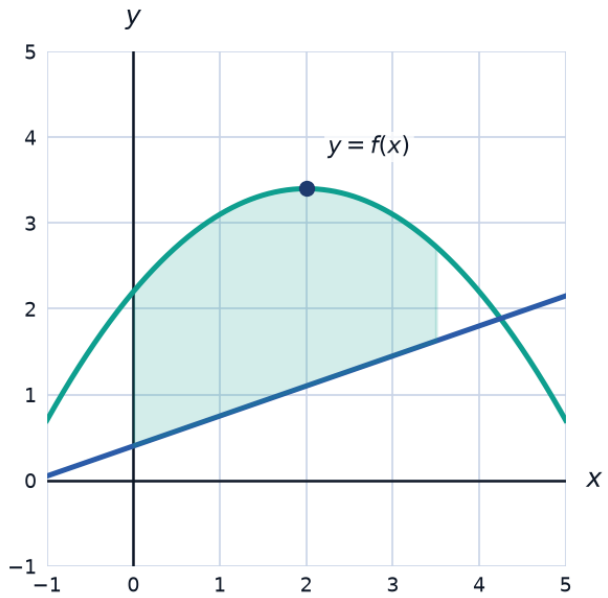
c. Converges to $\ln 2$

d. Inconclusive

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23 [Foundational] • Target: Integral Test and p-Series

Which conditions must $f(x)$ satisfy to apply the Integral Test to $\sum a_n$ where $a_n = f(n)$?



a. f must be continuous, positive, and decreasing on $[1, \infty)$

b. f must be differentiable and alternating

c. f must be bounded and periodic

d. $f'(x) > 0$ everywhere

24 [Foundational] • Target: Comparison Tests (Direct and Limit)

Determine the convergence of $\sum_{n=1}^{\infty} \frac{1}{n^3+5}$.

a. Converges by Direct Comparison with $\sum \frac{1}{n^3}$

b. Diverges by comparison with $1/n$

c. Inconclusive

d. Diverges by the Divergence Test

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25 [Foundational] • Target: Comparison Tests (Direct and Limit)

Use the Limit Comparison Test with $b_n = 1/n$ to test $\sum_{n=1}^{\infty} \frac{2n+1}{n^2-3}$.

a. Diverges because $L = 2 > 0$ and $\sum 1/n$ diverges

b. Converges because $L = 0$

c. Converges because $L = 2$

d. Inconclusive

26 [Foundational] • Target: Comparison Tests (Direct and Limit)

Which comparison series b_n is best suited for testing $\sum_{n=1}^{\infty} \frac{\sqrt{n}}{n^2+4n}$ via LCT?

a. $b_n = \frac{1}{n^{3/2}}$

b. $b_n = \frac{1}{n^2}$

c. $b_n = \frac{1}{n}$

d. $b_n = \frac{1}{\sqrt{n}}$

27 [Foundational] • Target: Alternating Series and Error Bounds

What are the two conditions of the Alternating Series Test (AST) for $\sum (-1)^n b_n$ ($b_n > 0$)?

a. $b_{n+1} \le b_n$ for all n and $\lim_{n\to\infty} b_n = 0$

b. $\sum b_n$ converges and $b_n > 0$

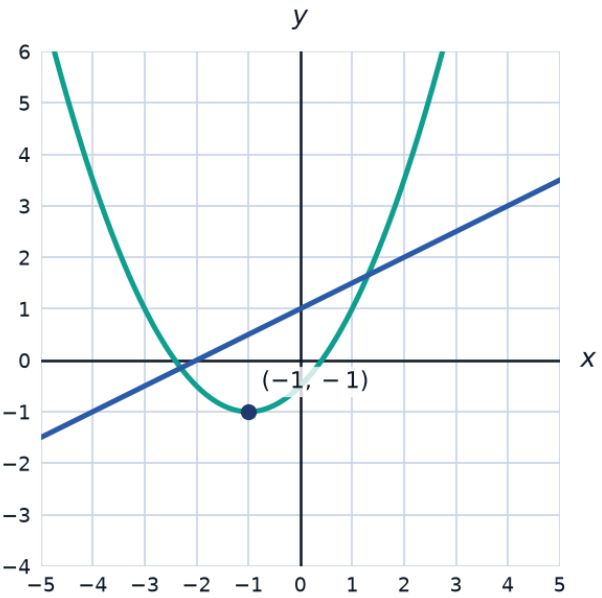
c. $\lim_{n\to\infty} b_n = 1$ and b_n is decreasing

d. b_n is geometric

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28 [Foundational] • Target: Alternating Series and Error Bounds

Classify $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n}$.



a. Conditionally convergent

b. Absolutely convergent

c. Divergent

d. Oscillating divergent

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Alternating Series and Error Bounds

If $S = \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^4}$ is approximated by S_3, what is the maximum error by the Alternating Series Estimation Theorem?

a. $|R_3| \leq \frac{1}{4^4} = \frac{1}{256}$

b. $|R_3| \leq \frac{1}{3^4} = \frac{1}{81}$

c. $|R_3| \leq \frac{1}{5^4}$

d. $|R_3| \leq \frac{1}{16}$

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30 [Intermediate] • Target: Ratio and Root Tests

State the conclusion of the Ratio Test when $L = \lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| < 1$.

a. The series converges absolutely

b. The series converges conditionally

c. The series diverges

d. The test is inconclusive

31 [Intermediate] • Target: Ratio and Root Tests

Apply the Ratio Test to $\sum_{n=1}^{\infty} \frac{3^n}{n!}$.

a. $L = 0 < 1 \Rightarrow$ Converges absolutely

b. $L = 3 > 1 \Rightarrow$ Diverges

c. $L = 1 \Rightarrow$ Inconclusive

d. $L = \infty \Rightarrow$ Diverges

32 [Intermediate] • Target: Ratio and Root Tests

Use the Root Test to determine the convergence of $\sum_{n=1}^{\infty} \left(\frac{2n+1}{3n-1}\right)^n$.

a. $L = 2/3 < 1 \Rightarrow$ Converges absolutely

b. $L = 2/3 \Rightarrow$ Diverges

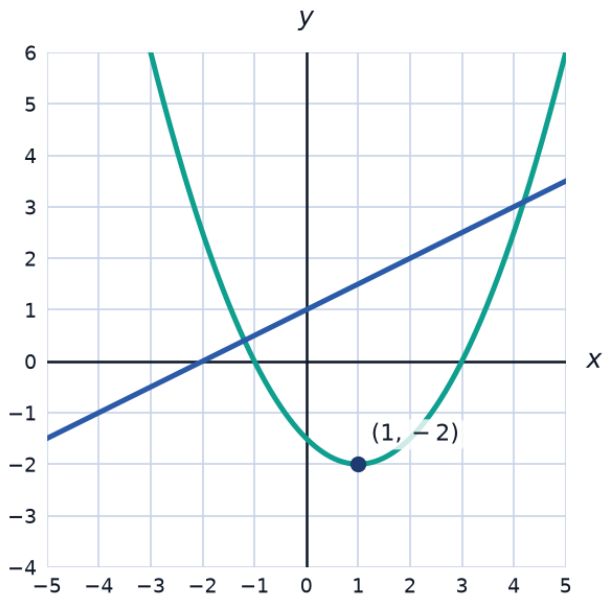
c. $L = 1 \Rightarrow$ Inconclusive

d. $L = 0$

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33 **[Intermediate]** • **Target: Alternating Series and Error Bounds**

By the alternating series estimation theorem, what is the fewest number of terms of $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n}$ needed to guarantee an error less than $\frac{1}{10000}$?



a. 10001

b. 9999

c. 20000

d. 10000

34 **[Intermediate]** • **Target: Ratio and Root Tests**

By the alternating series estimation theorem, what is the fewest number of terms of $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^2}$ needed to guarantee an error less than $\frac{1}{10000}$?

a. 101

b. 100

c. 99

d. 10000

35 **[Intermediate]** • **Target: The ratio and comparison tests**

By the alternating series estimation theorem, what is the fewest number of terms of $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n}$ needed to guarantee an error less than $\frac{1}{1000}$?

a. 1001

b. 999

c. 1000

d. 2000

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36 [Intermediate] • Target: Alternating series and the p-series test

By the alternating series estimation theorem, what is the fewest number of terms of $\sum_{n=0}^{\infty} \frac{(-1)^n}{n!}$ needed to guarantee an error less than $\frac{1}{10000}$?

a. 8

b. 9

c. 7

d. 10000

37 [Intermediate] • Target: The integral test

By the alternating series estimation theorem, what is the fewest number of terms of $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^2}$ needed to guarantee an error less than $\frac{1}{1000}$?

a. 31

b. 32

c. 30

d. 1000

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Choosing the right test • Cross-Lesson: 8.02 Volumes of Revolution

By the alternating series estimation theorem, what is the fewest number of terms of $\sum_{n=0}^{\infty} \frac{(-1)^n}{n!}$ needed to guarantee an error less than $\frac{1}{1000}$?

a. 8

b. 6

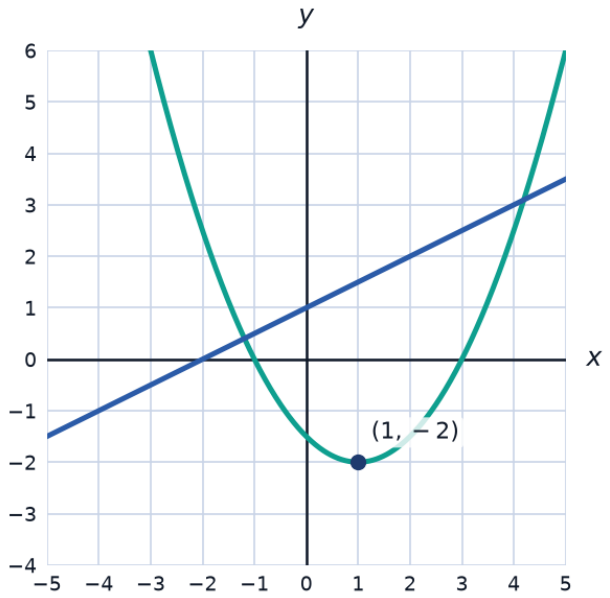
c. 7

d. 1000

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39 [Challenging] • Target: The limit comparison test • Cross-Lesson: 8.03 Arc Length and Improper Integrals

By the alternating series estimation theorem, what is the fewest number of terms of $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^2}$ needed to guarantee an error less than $\frac{1}{100}$?



a. 11

b. 9

c. 100

d. 10

40 [Challenging] • Target: The root test • Cross-Lesson: 9.01 Separable Differential Equations

By the alternating series estimation theorem, what is the fewest number of terms of $\sum_{n=0}^{\infty} \frac{(-1)^n}{n!}$ needed to guarantee an error less than $\frac{1}{100}$?

a. 6

b. 4

c. 100

d. 5

41 [Challenging] • Target: Absolute versus conditional convergence • Cross-Lesson: 1.01 Precalculus Review for Calculus

Series $\sum_{n=1}^{\infty} (-1)^{n-1}/\sqrt{n}$ converges by alternating series test. Is it absolutely convergent?

a. Yes, $p = 1/2 < 1$ diverges

b. No, $\sum 1/\sqrt{n}$ diverges ($p = 1/2 \leq 1$)

c. Yes, $\sum 1/\sqrt{n}$ converges

d. Cannot determine

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42 [Challenging] • Target: Estimating the sum of an alternating series • Cross-Lesson: 10.01 Calculus with Parametric and Polar Curves

Integral test convergence of $\sum_{n=1}^{\infty} 1/n(\ln n)^2$ (from $n=2$) gives sum approximately:

a. converges

b. diverges

c. equals e

d. equals π^2

43 [Challenging] • Target: A combined convergence-test challenge • Cross-Lesson: 11.01 Sequences and Series

Which test determines $\sum n^2/e^n$? Answer: by ratio test $L =$

a. e

b. 1

c. $1/e < 1$

d. e^2

44 [Challenging] • Target: A multi-step convergence investigation • Cross-Lesson: 12.02 Power Series and Taylor Series

Series $\sum_{n=1}^{\infty} n \cdot (1/2)^n$: ratio test gives $L =$

a. 1

b. 2

c. $1/2$

d. $n/2$

45 [Challenging] • Target: Integral Test and p-Series • Cross-Lesson: 13.01 Vectors and Vector-Valued Functions

The ratio test for $\sum_{n=0}^{\infty} x^n/n!$: $|a_{n+1}/a_n| = |x|/(n+1) \to$

a. $|x|$

b. 1

c. 0

d. ∞

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